

SYMBOLIC MICROINSTRUCTIONS

- Symbols are used in microinstructions as in assembly language
- A symbolic microprogram can be translated into its binary equivalent by a microprogram assembler.

Sample Format

five fields: label; micro-ops; CD; BR; AD

Label: may be empty or may specify a symbolic
 address terminated with a colon

Micro-ops: consists of one, two, or three symbols separated by commas.

CD: one of {U, I, S, Z}, where

- U: Unconditional Branch
- I: Indirect address bit
- S: Sign of AC
- Z: Zero value in AC

BR: one of {JMP, CALL, RET, MAP}

AD: one of {Symbolic address, NEXT, empty}



SYMBOLIC MICROPROGRAM - FETCH ROUTINE

During FETCH, Read an instruction from memory and decode the instruction and update PC

Sequence of microoperations in the fetch cycle:

```
AR ← PC
DR ← M[AR], PC ← PC + 1
AR ← DR(0-10), CAR(2-5) ← DR(11-14), CAR(0,1,6) ← 0
```

Symbolic microprogram for the fetch cycle:

```

      ORG 64
FETCH: PCTAR      U JMP NEXT
      READ, INCPC U JMP NEXT
      DRTAR      U MAP
  
```

Binary equivalents translated by an assembler

Binary address	F1	F2	F3	CD	BR	AD
1000000	110	000	000	00	00	1000001
1000001	000	100	101	00	00	1000010
1000010	101	000	000	00	11	0000000

